

The MOND acceleration scale as dark-sector pressure

a derived $a_0(z)$ that switches MOND off at recombination

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Abstract. In relativistic completions of Milgrom's modified dynamics the acceleration scale a_0 is inserted as a

constant of the theory and fitted to galaxy data. We observe that in the Aether Scalar Tensor theory (AeST) of Skordis

and Złośnik — currently the only MOND-like relativistic theory that reproduces the cosmic microwave background —

the dark sector already supplies a pressure, and that identifying $\mathcal{A}(Q) \equiv a_0^2(Q) = \kappa^2 G[-K(Q)]$ removes a_0 as an

independent constant. Two consequences follow without further adjustment. First, a_0 is fixed by the dark-energy

density, $a_0 = \kappa c \sqrt{(G\rho_\Lambda)}$, giving $9.36 \times 10^{-11} \text{ m s}^{-2}$ for $\kappa = 1/2$; we emphasise that κ here is *measured*, $\kappa = 0.551 \pm 0.043$,

and is *not* derived — four candidate values lie within 2σ . Second, and independently of κ , the dark-sector pressure

evolves, so a_0 must evolve with it in a form fixed by the same equation: $a_0^2(z)/a_0^2(0) = \sqrt{(1+v_0^2)} / \sqrt{(1+v_0^2(1+z)^6)}$,

which rises monotonically to a maximum today and gives $a_0(z=1090)/a_0(0) = 6.0 \times 10^{-3}$. MOND is therefore absent

when the CMB is imprinted, making the CMB pass structural rather than fitted. That a_0 may depend on the

cosmological scalar-field value, and hence evolve, is not new — it was computed for TeVeS by Bekenstein and Sagi,

who obtained a_0 *decreasing* with time; the identification with the pressure specifically, the opposite sign of evolution,

and the AeST setting are what is new here. We report what the construction passes ($c_T = 1$ exactly, no ghost, $\gamma_{PPN} = 1$,

a CLASS CMB fit, SPARC rotation curves at 0.108 dex, the baryonic Tully–Fisher relation as a theorem, weak lensing

from 40 kpc to 2.2 Mpc, and the solar system), and we state plainly what it does not: galaxy clusters remain short by a

factor 2.08 at R_{500} , and we prove that this construction cannot supply the deficit, because the available amplitude

reduces to a charge-abundance ratio that is 10^2 – 10^3 too small even if the dark sector's conserved charge were the

entire dark matter content of the universe.

1. Introduction

Rotation curves of disc galaxies do not fall in the way the visible baryons require. Milgrom's modified dynamics [1] accounts for this with remarkable economy: given the baryonic mass distribution it predicts the rotation curve across five decades in mass, and the radial acceleration relation [2] is tight enough that the scatter is comparable to the observational uncertainties. The construction contains a single new constant, an acceleration $a_0 \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$, above which dynamics is Newtonian and below which it is not.

The numerical proximity of a_0 to accelerations built from cosmological quantities — cH_0 , or equivalently $c\nu/(G\rho_\Lambda)$ — has been remarked on since the earliest MOND papers and is discussed at length in the review literature [3]. It has nonetheless remained a coincidence: in every working formulation a_0 is a number fitted to galaxies, and its resemblance to cosmology carries no consequence. Milgrom has proposed specific normalisations, most recently with coefficient $1/2\pi$ [4].

Relativistic completions have been harder. Bekenstein's TeVeS [5] reproduced the phenomenology but did not survive the cosmic microwave background. The situation changed with the Aether Scalar Tensor theory (AeST) of Skordis and Złośnik [6], which recovers MOND in quasi-static systems *and* fits the CMB and the matter power spectrum. AeST is the setting for everything below, and the results of this paper are consequences of their theory rather than of a new one; in their formulation a_0 remains an inserted constant.

The observation of this paper is that AeST's dark sector already contains a quantity with the right dimensions to supply a_0 , namely its own pressure, and that identifying the two removes a free constant and forces an evolution law.

2. The identification

We take AeST in the form of Ref. [6]: a metric $g_{\mu\nu}$, a unit-timelike vector A^μ with $A^\mu A_\mu = -1$, and a scalar φ , with the invariants

$$Y = q^{\mu\nu} \nabla_\mu \varphi \nabla_\nu \varphi, \quad Q = A^\mu \nabla_\mu \varphi,$$

where $q^{\mu\nu} = g^{\mu\nu} + A^{\mu}A^{\nu}$. The dark sector enters through a function $\mathcal{F}(Y,Q)$ whose Q -dependent part $K(Q)$ carries the pressure, $p = K$, and whose Y -dependent part supplies the MOND limit with a scale $\mathcal{A} = a_0^2$ inserted by hand.

We instead set

$$\mathcal{A}(Q) = a_0^2(Q) = \kappa^2 G [-K(Q)] \quad (1)$$

with κ dimensionless. In words: the MOND scale is the dark sector's pressure, expressed as an acceleration. Nothing is added to the theory — $K(Q)$ was already present, doing another job — and one constant is removed.

Two remarks fix the interpretation. Since $w = -1$ requires $p = -\rho$ and the sector's exact thermodynamics give $n = K'$, $\rho = QK' - K$, $p = K$, the background sits near $K' = 0$; we return in Sec. 4 to the fact that it sits *near* rather than *at* that point, which is what makes the evolution law non-trivial. And because Eq. (1) ties $\mathcal{A}$ to the same object that drives cosmic acceleration, the value of a_0 becomes a statement about ρ_Λ .

3. First consequence: the scale is fixed, the coefficient is measured

With $-K = \rho_\Lambda c^2$ on the background, Eq. (1) gives

$$a_0 = \kappa c \sqrt{G\rho_\Lambda} = cH_\Lambda/Z, \quad Z \equiv \sqrt{(8\pi/3)}/\kappa, \quad (2)$$

the second form following because the Friedmann equation supplies $H_\Lambda = \sqrt{(8\pi/3)} \cdot \sqrt{G\rho_\Lambda}$. For $\kappa = 1/2$, equivalently $Z = 2\sqrt{(8\pi/3)} = \sqrt{(32\pi/3)} = 5.78881$, this yields $a_0 = 9.3619 \times 10^{-11} \text{ m s}^{-2}$. Using ρ_{total} and cH_0 in place of ρ_Λ and cH_Λ — a defensible alternative bookkeeping which we carry throughout as the "alt" footing — gives $1.1279 \times 10^{-10} \text{ m s}^{-2}$ instead. We quote both for every dimensionful result.

The coefficient is not derived.

We state this in the strongest terms because the literature is full of near-misses. A distance-free determination from the radial acceleration relation gives $\kappa = 0.551 \pm 0.043$, and the baryonic Tully–Fisher route gives 0.465 ± 0.076 ; $\kappa = 1/2$ lies 1.19σ from the former, and at least four candidate values sit within 2σ of one another. The data are therefore consistent with $\kappa = 1/2$ but do not select it.

Furthermore, the apparent depth of the question is illusory. Because the π in Eq. (2) cancels between $\sqrt{(8\pi/3)}$ and Z , the statement $\kappa = 1/2$ is identically the statement $Z = 2\sqrt{(8\pi/3)}$, i.e. that Z is exactly twice the Friedmann factor. Any normalisation argument that returns $\kappa^2 = 8\pi\varepsilon$ and then reports $\varepsilon = 1/32\pi$ has not derived anything: the 8π is carried over from its own left-hand side, and the residual 4 in 32π is κ^{-2} . The entire open content is one factor of two, and we do not claim to have it. By contrast, the coefficient is not what the next section rests on.

4. Second consequence: a derived $a_0(z)$

The dark-sector pressure evolves, so Eq. (1) forces a_0 to evolve. Writing $u = Q - Q_0$ and $v = \mu_p u / M^2$ for the sector's dimensionless displacement, with the brane form $K(u) = -M^4 \sqrt{1 - \mu_p^2 u^2 / M^4}$ at $\beta \equiv \mu_p^2 \Lambda_D^2 / M^4 = 1$, and using $n \propto a^{-3}$ so that $v(z) = v_0(1+z)^3$, Eq. (1) gives

$$a_0^2(z)/a_0^2(0) = \sqrt{(1+v_0^2)} / \sqrt{(1+v_0^2(1+z)^6)} \quad (3)$$

with a single parameter v_0 . Constraints already carried by the sector confine $v_0 \in [2.14, 17.7] \times 10^{-5}$, whence a transition redshift $z_t = v_0^{-1/3} - 1 \in [17, 35]$ and, at recombination, $a_0(z=1090)/a_0(0) = 6.0 \times 10^{-3}$.

Three features deserve emphasis. (i) Eq. (3) is *monotonically rising* toward the present: a_0 is at its maximum today. (ii) It is exactly κ -independent, so it tests the identification (1) without depending on the unresolved coefficient. (iii) MOND is therefore *off* when the CMB is imprinted. This last point is the reason the CMB is not a struggle here: the clustering behaviour required at $z \sim 10^3$ is a consequence of Eq. (3) rather than an assumption, and the failure mode that removed TeVeS from contention does not arise.

Note also that $v_0 \neq 0$ is required for Eq. (3) to be non-empty, so the background sits at $u_0 = v_0 \Lambda_D \neq 0$ and $w = -1$ holds only to $\mathcal{O}(v_0^2)$, i.e. to a fractional offset in $[4.6 \times 10^{-10}, 3.1 \times 10^{-8}]$. That is far below any current determination of w_0 , but it should be written as $w = -1 + \mathcal{O}(v_0^2)$ rather than as exactly -1 .

Relation to prior work.

That a_0 may depend on the cosmological value of the dark-sector scalar, and hence evolve, is *not* new. Bekenstein and Sagi computed exactly this for TeVeS [7], finding a_0 varying as the exponential of the scalar field and *decreasing* with time; Ref. [3] treats the general statement — that a_0 for a quasi-static system depends on the field value at the epoch the system collapsed — as established background. What is specific to Eq. (1) is the identification with the *pressure* rather than with an exponential of the field, so that no new function is introduced; the *opposite sign* of the resulting evolution; and the AeST setting, which is the one that survives the CMB. We make no priority claim beyond that, and we have not performed an exhaustive literature search.

5. What the construction passes

All fitted numbers below use the algebraic interpolation $g^2 = g_{bar}^2 + a_0 g_{bar}$, equivalently Milgrom's $v(y) = \sqrt{(1+1/y)}$ [8], unless stated otherwise.

- **Tensor sector.** $c_T = 1$ exactly, as in the parent theory, so GW170817 is satisfied identically. The no-ghost condition of the parent theory transfers: with the promotion the two-field longitudinal conditions reduce to the two standard AQUAL conditions.
- **Lensing.** $\gamma_{PPN} = 1$. A fit to the weak-lensing radial acceleration relation of Ref. [9] from 40 kpc to 2.2 Mpc, with no free amplitude, gives $\chi^2/\text{dof} = 2.03$ (canonical footing) and 0.94 (alt).

- **CMB.** A CLASS run with Eq. (3) in place returns $\Delta\chi^2 = 1.34$ over 4998 multipoles relative to the unmodified fit, with θ^* displaced by 0.18 of the Planck uncertainty.
- **Galaxies.** SPARC rotation curves at 0.108 dex scatter with stellar mass-to-light ratio $Y = 0.70$. The baryonic Tully–Fisher relation $v^4 = GM_{\text{ba0}}$ follows as a *theorem* of the interpolation rather than as a fit, since r drops out identically, with $d \ln v / d \ln a_0 = 1/4$.
- **Solar system.** The algebraic interpolation above approaches Newtonian dynamics as $g_{\text{bar}} + a_0/2$, i.e. with a constant residual anomaly, which exceeds planetary ranging bounds by a factor $\sim 1.3 \times 10^3$. It must therefore be read as an effective description valid where the data lie, with the exponential form $v(y) = [1 - e^{-\sqrt{y}}]^{-1}$ of Ref. [2] as the interpolation proper; that form approaches Newtonian exponentially and the anomaly at Earth is unmeasurably small. The two differ by at most 0.057 dex over $g_{\text{bar}}/a_0 \in [0.03, 10]$, i.e. half the SPARC scatter, so the fits above are unaffected; but every quoted number is interpolation-conditional and we flag it as such.

6. What it does not pass, and a no-go for repairing it

Galaxy clusters.

Taking the baryons of the twelve X-COP clusters [10, 11] and predicting the dynamical mass from the interpolation above, the median discrepancy is a factor 2.08 at R_{500} (canonical footing; 1.92 alt), rising to 4.21 inside $0.1R_{500}$. This is the long-standing cluster problem of MOND-like theories [3]; it is not specific to Eq. (1) and it is not solved by AeST as published.

And Eq. (1) cannot repair it.

The natural repair is to let the local dark-sector charge modify the effective interpolation, since $\mathcal{A}$ is now Q -dependent. Expanding, and using $\mathcal{A}' = \kappa^2 G(-K') = -\kappa^2 G n$ together with $-K = \rho \Lambda c^2$ and $\rho_{\text{charge}} = Q_0 n$, the induced correction is

$$\Delta\mu = -(\rho_{\text{charge}} / \rho \Lambda c^2) \Phi T(y), \quad T = y \mathcal{F}''(y), \quad (4)$$

in which v_0 , Λ_D , Q_0 , M and κ all cancel: the amplitude is a pure charge-abundance ratio. Reproducing the cluster deficit requires a coefficient of order 0.5; Eq. (4) supplies at most $\sim 10^{-9}$. Even setting the dark sector's conserved charge to the *entire* dark matter content, $\Omega_{\text{kd}} = \Omega_{\text{dm}}$, leaves the term short by a factor 10^2 – 10^3 ; matching would require $\Omega_{\text{kd}} \sim 2.7 \times 10^2$. The charge is a trace species, and it is one for the same reason Eq. (3) switches MOND off early. We regard this as a genuine no-go for this particular repair, not as a bound to be improved.

Two further open problems.

The sector's pressureless excitation, which the CMB fit requires at the level of Ω_{dm} , is forced cold by that same fit and has no evident mechanism halting its collapse in galaxies; followed through, the endpoint is inconsistent with the observed mass of Sgr A* by some five orders of magnitude. And κ remains measured rather than derived, as stated in Sec. 3. We list these rather than defer them.

7. Falsifiable predictions

1. **No a_0 evolution below $z \sim 5$.** Eq. (3) is flat to $\sim 10^{-8}$ at low redshift. Any robust detection of a_0 evolution below $z \sim 5$ falsifies it. Current intermediate-redshift determinations of the radial acceleration relation are in mild tension with this and constitute the sharpest near-term test.
2. **Suppressed rotation velocities at cosmic dawn.** Since $v^4 = GM_{ba_0}$ is a theorem, $v(z)/v(0) = [a_0(z)/a_0(0)]^{1/4}$, giving deficits of 0.34%, 7.8% and 18.9% at $z = 10, 20$ and 30 . The effect becomes detectable near $z \approx 17$.
3. **Wide binaries.** The framework predicts a velocity-boost ratio $\gamma_v = 1.2139$ (canonical) or 1.2592 (alt) in the Gaia DR4 wide-binary sample, against 1 for Newtonian dynamics; the two footings are separated by 2.68σ , so DR4 distinguishes them. This prediction was registered before the data.

8. Conclusion

Identifying the MOND acceleration scale with the dark sector's pressure, Eq. (1), removes a_0 as an independent constant of AeST and forces an evolution law, Eq. (3), which is κ -independent, rises to a maximum today, and puts a_0 at 0.6% of its present value at recombination. The CMB pass of the parent theory thereby becomes structural. The construction retains $c_T = 1$, no ghost, $\gamma_{PPN} = 1$, the SPARC relation at 0.108 dex, the Tully–Fisher relation as a theorem, and weak lensing to 2.2 Mpc. It does not solve galaxy clusters, and we have shown that it cannot: the available amplitude is a charge-abundance ratio too small by 10^2 – 10^3 . The coefficient κ is measured, not derived, and the open content of that question is a single factor of two.

Acknowledgments

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